Bloomberg, Capital IQ & Advanced Modeling — Hands-On

Hands-On Drills — Do the Task, Not Just Define It (India, 2026)

Step-by-step functions, T-codes, worked models and interview drills

Contents

Terminal Orientation: Bloomberg & Capital IQ Function Drills

What you'll be able to do

Sit down at a Bloomberg terminal cold and drive it: pick the right asset class with the yellow keys, load a ticker, run the core functions an analyst uses every day (DES, FA, GP, EQS, RV, BICS, WACC, BETA), and read the screens fast enough to answer a "what's this company worth and how's it trading" question in five minutes. You'll also know where the same things live in S&P Capital IQ (CapIQ) so you can switch platforms without re-learning. By the end you can look up Infosys, read its description, financials and price chart, and pull a peer set — the raw material for every model in this guide.

The drill — step by step

1. The yellow keys (market sectors). The keyboard's yellow keys set the asset class so the terminal knows what "INFO" means. The ones you'll touch: `EQUITY` (stocks), `GOVT` (sovereign bonds), `CORP` (corporate bonds), `CMDTY` (commodities), `CRNCY` (FX), `INDEX`, `MMKT` (money markets), `MTGE`. The workflow is always: **type the name → press the yellow key → press `<GO>`.**

`<GO>` (the green key) executes. `<HELP>` once explains the current screen; `<HELP><HELP>` (twice) opens live chat with the Bloomberg help desk — genuinely useful, a real person answers.

2. Load a security. Type `INFO` then press `EQUITY`. The autocomplete/ticker-lookup drops a match list. Infosys trades in several places, so you disambiguate by exchange code:

What you type	Meaning
`INFO IN <EQUITY> <GO>`	Infosys, NSE India listing (`IN` = India)
`INFO US <EQUITY> <GO>`	Infosys ADR on NYSE
`INFY IN <EQUITY>`	also resolves (ticker vs. short name)

`IN` is the Bloomberg country/exchange code for India. If you don't know the ticker, type the company name and hit `EQUITY <GO>` — the security-finder (SECF) list appears; arrow down and Enter.

3. Now run the core functions. Once a security is loaded, you just type the mnemonic and `<GO>`; it applies to the loaded ticker. Drill them in this order:

- **`DES <GO>`** — Description. Business summary, sector, key execs, listing details, GICS/BICS classification, a snapshot of price and market cap. Your first stop on any name.
- **`FA <GO>`** — Financial Analysis. The workhorse. Tabs across the top: Income Statement (IS), Balance Sheet (BS), Cash Flow (CF), Ratios, Segments, Multiples. Change the period toggle to Annual/Quarterly and the currency (INR/USD). This is where you read 5 years of P&L.
- **`GP <GO>`** — Graph Price. Candlestick/line price chart. Change the range with the field boxes (`1D`, `1M`, `1Y`, `5Y`). `GIP` = intraday, `GPO` = price only.
- **`GPO`, `HP`** — `HP` (Historical Prices) is the tabular price history you'd export.

- `EQS <GO>` — Equity Screening. Build a peer/universe screen with criteria (country = India, BICS sub-industry = IT Consulting & Services, market cap > $5bn). Returns a list you can save — this seeds your comps set.
- `RV <GO>` — Relative Value. Auto-builds a comparable-companies grid for the loaded name (EV/EBITDA, P/E, growth) against a peer group Bloomberg guesses; you can edit the peers. This is a pre-baked comps table.
- `BICS` — Bloomberg Industry Classification. Used inside EQS/RV to define the peer universe (Bloomberg's answer to GICS).
- `WACC <GO>` — Weighted Average Cost of Capital. Bloomberg's computed cost of equity (via CAPM), cost of debt, capital weights, and blended WACC. Read it, don't trust it blindly — check the inputs.
- `BETA <GO>` — regression of the stock vs. its index (adjusted/raw beta, R^2). Set the index to NSE Nifty for an Indian name.
- `CN <GO>` — Company News. `TOP <GO>` = top market news.
- `PORT <GO>` — Portfolio & Risk Analytics (needs a saved portfolio).
- `FLDS <GO>` — field finder; searches every data field name (needed later for Excel BDP).

Navigation muscle memory: the **red** `<CANCEL>` stops a running query; `<MENU>` (or the left-arrow) goes back one screen; `PANEL` **/green arrows** cycle between the four screen panels; `1<GO>` ... `4<GO>` open a function on a numbered panel.

4. Capital IQ (the cheaper, web-based cousin). CapIQ is browser-based, not a keyboard terminal. Layout is a left-nav of modules:

- **Companies** — search a company → a company profile with tabs: Tearsheet, Financials, Comparable Companies, Transactions, Estimates, Ownership.
- **Screening** — the EQS equivalent; build company/transaction screens with a criteria builder.
- **Comps** — pre-built comparable-company templates you clone.
- **Transactions** — M&A/precedent-deal database (used in Chapter 4).
- **Tearsheet** — a one-page company summary (description, financials, multiples) you can export to PDF/Excel — the standard "hand this to the MD" page.

Same job, different door: Bloomberg `DES` ≈ CapIQ company profile; `FA` ≈ CapIQ Financials tab; `RV` / `EQS` ≈ CapIQ Comps/Screening; deal work ≈ CapIQ Transactions.

Worked example — Infosys in five screens: 1. `INFO IN EQUITY <GO>` → loads. 2. `DES <GO>` → read: "Infosys Ltd, IT consulting, HQ Bengaluru, mkt cap ≈ ₹6.5 lakh cr / ~$78bn, ADR: INFY US." 3. `FA <GO>` → IS tab, Annual, INR → note FY revenue ≈ ₹1.53 lakh cr, EBIT margin ≈ 21%. 4. `GP <GO>`, set 5Y → see the multi-year price trend. 5. `RV <GO>` → peer grid vs. TCS, Wipro, HCLTech — read Infosys EV/EBITDA vs. peer median.

The output

Your "orientation output" is a filled scratch sheet — one row per function, so you can answer a snap valuation question:

Function	Field read	Infosys value (illustrative)
DES	Sector / mkt cap	IT Services / ~$78bn
FA (IS)	Revenue / EBIT margin	₹1.53 lakh cr / ~21%
FA (Ratios)	ROE	~30%
GP 5Y	Trend	up, cyclical dips
BETA	Adj. beta vs Nifty	~0.7
WACC	Blended WACC	~11–12%
RV	EV/EBITDA vs peer median	e.g. 18x vs 16x → slight premium

Checks & gotchas

- **Wrong yellow key = wrong or no security.** `INFO` alone means nothing; `INFO EQUITY` does. Always press the sector key.
- **Which listing?** ADR (`US`) vs. local (`IN`) have different currency, price and multiples. State which one you're using.
- **Currency & scale on FA.** Toggle INR vs. USD deliberately; Indian filings are often in ₹ crore — confirm the units box before you quote a number.
- **WACC/BETA are model outputs, not gospel.** Bloomberg's beta window and index choice drive the number — check the regression index is Nifty, not S&P 500, for an Indian name.
- **RV peers are auto-picked.** Bloomberg's default peer set will include names you'd reject; you must curate (Chapter 3).

Interview drill

Q: "Walk me through the first three things you do on a Bloomberg when handed a new ticker." A: "Load it with the right yellow key, run `DES` to understand the business, size and classification, then `FA` to read the last few years of income statement, margins and returns. `GP` for the price trend. That tells me what it does, how it earns and how the market's treating it before I touch a model."

Q: "What's the difference between EQS and RV?" A: " `EQS` is a screener — I define criteria and it returns a universe. `RV` is relative-value — it takes the loaded name and builds a comps grid against a peer group, showing valuation multiples side by side. EQS finds the peers; RV compares them. In CapIQ these are the Screening and Comparable Companies modules."

Q: "Bloomberg vs. Capital IQ — when would you reach for which?" A: "Bloomberg for live markets, fixed income, FX, real-time news and speed once you know the mnemonics. CapIQ for deep private-company data, M&A/transaction comps and clean Excel-exportable tearsheets. Most desks use both."

Practise free

You can rehearse 80% of this without a terminal: - **Bloomberg Market Concepts (BMC)** — Bloomberg's own ~8-hour paid-but-cheap e-course (often free via a college/library licence); teaches the mental model and mnemonics. - **Screener.in** — India's best free `FA` substitute: 10-year P&L, balance sheet, cash flow, ratios per company. Look up Infosys and read exactly what FA shows. - **Tijori Finance** — free tier gives segment and peer views (an RV-style comps grid). - **NSE India / BSE India** — official price history and corporate filings (your `HP` / `CN`). - **TIKR / stockanalysis.com** — free CapIQ-style tearsheets and global comps. Drill: open Screener.in, pull Infosys, TCS, Wipro, HCLTech side by side, and reproduce the RV table above by hand. That single exercise builds the reading speed the terminal only accelerates.

Pulling & Cleaning Data into Excel (BDP/BDH/BQL, CapIQ Plugin)

What you'll be able to do

Get data out of Bloomberg and Capital IQ into a spreadsheet **as live, refreshable formulas** — not copy-paste that rots the moment the price moves. You'll write `=BDP` for a single current value, `=BDH` for a time series, `=BDS` for bulk/list data, know when to reach for BQL, and pull the same financials from CapIQ with `=CIQ` formulas. You'll build a clean, model-ready "Data" tab: one place all raw pulls land, from which your model reads — the discipline that stops a model from silently breaking. Worked outcome: five years of Infosys financials sitting in a tab your DCF/comps can point at.

The drill — step by step

0. Install & log in. The Bloomberg add-in installs from the terminal (`WAPI` / `API<GO>` → "Excel Add-in") and needs you logged into Bloomberg on that machine. In Excel you'll see a **Bloomberg** ribbon tab. CapIQ's plugin installs separately and adds a **Capital IQ** ribbon (log in with your S&P credentials).

1. `=BDP` — Bloomberg Data Point (one static/current value). Syntax: `=BDP("security", "field")`.

```
=BDP("INFO IN Equity","PX_LAST")          → last price
=BDP("INFO IN Equity","CUR_MKT_CAP")      → market cap
=BDP("INFO IN Equity","BEST_PE_RATIO")    → forward P/E (BEST = consensus estimate)
=BDP("INFO IN Equity","CRNCY")            → INR
```

The trick is knowing the **field mnemonic**. Find them with `FLDS <GO>` on the terminal (searchable field dictionary) or the FieldSearch button on the ribbon. `PX_LAST`, `CUR_MKT_CAP`, `SALES_REV_TURN`, `EBITDA`, `IS_EPS`, `BS_TOT_ASSET`, `TRAIL_12M_…`, `BEST_…` (estimates) are the ones you'll reuse.

2. `=BDH` — Bloomberg Data History (a time series). Syntax: `=BDH("security","field(s)","start","end", [optional args])`.

```
=BDH("INFO IN Equity","PX_LAST","1/1/2021","12/31/2025","Per=M")
```

`Per=M` (monthly), `Per=Q`, `Per=Y` set frequency; `Fill=P` carries prices over non-trading days; `Days=W` restricts to weekdays. BDH spills a two-column (date, value) array **downward** — leave blank cells below the formula or it errors with `#N/A Requesting Data` colliding into your labels.

3. `=BDS` — Bloomberg Data Set (bulk / list data that returns many rows/cols). Used for things that aren't a single number: index members, dividend history, the full list of a company's segments or peers.

```
=BDS("NIFTY Index","INDX_MEMBERS")     → all Nifty constituents
=BDS("INFO IN Equity","DVD_HIST")      → dividend history table
```

4. BQL — Bloomberg Query Language (the modern, powerful way). BQL does filtering, aggregation and universe screening in one formula, returning a table. It's what you graduate to when BDP/BDH get clumsy.

Conceptually: `get( … )` fields `for( … )` a universe `with( … )` parameters.

```
=BQL("members('NIFTY Index')","name, px_last, pe_ratio")
```

returns every Nifty member with its name, price and P/E in one spill. You don't need mastery on day one — know it exists and that it replaces stacking 50 BDP calls.

5. CapIQ Excel plugin — `=CIQ` formulas. Same idea, S&P's data. `=CIQ("IQ ticker / identifier","mnemonic", [period])`.

```
=CIQ("INFO:","IQ_TOTAL_REV","IQ_FY-4")     → revenue, 4 years ago
=CIQ("INFO:","IQ_EBITDA","IQ_FY")          → latest FY EBITDA
=CIQ("INFO:","IQ_EBITDA","IQ_FY-1")        → prior FY
```

`IQ_FY`, `IQ_FY-1…-4` walk back fiscal years; `IQ_FQ` for quarters; `IQ_NTM` / `IQ_FY+1` for forward estimates. CapIQ mnemonics (`IQ_TOTAL_REV`, `IQ_EBIT`, `IQ_NI`, `IQ_TOTAL_DEBT`, `IQ_CASH_ST_INVEST`, `IQ_TEV` = enterprise value) come from the plugin's formula builder — use it, don't memorise.

6. Build the clean Data tab. Discipline that separates a pro model from a mess: - One tab named `Data_Raw`. Formulas only reference it; the model never calls BDP directly in the middle of a calc. - **Columns = periods (FY-4…FY), rows = line items.** Put the entity ticker in one cell (`B1`) and reference it: `=CIQ($B$1,"IQ_TOTAL_REV","IQ_FY-4")`. Change one cell → whole tab re-points to a new company. - Immediately below each live-formula block, **paste-special → values** into a mirror block if you need a point-in-time snapshot (see gotchas). - Add a **units row** and a **currency cell**; convert to a single reporting currency explicitly.

Worked example — Infosys 5-yr revenue & EBITDA, model-ready:

Row / `IQ_FY-`	FY-4	FY-3	FY-2	FY-1	FY
Formula (Revenue)	`=CIQ($B$1,"IQ_TOTAL_REV","IQ_FY-4")`	… `-3`	… `-2`	… `-1`	… `IQ_FY`
Revenue (₹ cr)	100,472	121,641	146,767	153,670	162,990
EBITDA (₹ cr)	26,000	30,700	34,900	36,100	38,900
EBITDA margin	25.9%	25.2%	23.8%	23.5%	23.9%

(Numbers illustrative — the point is every cell is a formula keyed to `$B$1`.)

The output

A single `Data_Raw` tab: ticker cell up top, a currency/units header, one clean block per statement (IS, BS, CF), periods across, live formulas that refresh on open, and a values-mirror for the version you actually valued off. Your comps and DCF tabs contain **zero** BDP/CIQ calls — they read from `Data_Raw` by cell reference. Hand it to a colleague, they change `$B$1` from `INFO:` to `TCS:` and the whole book reprices.

Checks & gotchas

- **`#N/A Requesting Data` / `#N/A Invalid Field`** — data still loading (wait/refresh) vs. a wrong mnemonic (fix via FLDS/formula builder). Don't build on top of a cell that's still `Requesting`.
- **Point-in-time vs. restated.** Live formulas pull *today's* view of history — companies restate, and consensus estimates drift daily. If your valuation must be reproducible, paste-special values and date-stamp the snapshot. This is the single most common "my model changed overnight" bug.
- **Currency & scale.** CapIQ may return USD while Bloomberg FA showed INR; ₹ crore vs. ₹ mn vs. absolute. Force one currency, one scale, and label it. A 100x error hides here.
- **Fiscal-year misalignment.** `IQ_FY` for an Indian March year-end ≠ a US December year-end peer — you'll calendarise in Chapter 3.
- **Blank-cell collisions.** BDH/BDS spill arrays; leave room or they overwrite (or error against) adjacent cells.
- **Broken on a machine without a login.** These formulas only resolve where the add-in + entitlement live. Email the workbook and the recipient sees `#NAME?` unless they too have the plugin — send the values-mirror.

Interview drill

Q: "Difference between BDP, BDH and BDS?" A: "BDP returns one current/static value for one field. BDH returns a historical time series with a frequency you set. BDS returns bulk or list-type data — index members, dividend schedules — that isn't a single scalar. One point, one history, one set."

Q: "How do you make sure a Bloomberg-linked model is reproducible?" A: "Isolate all live pulls in a Data tab, then paste-special the values you actually valued off and date-stamp them, because live formulas re-pull restated financials and shifting consensus every time the book opens. The audit version is static; the live version is for updating."

Q: "Why keep a separate Data tab instead of calling BDP inside the model?" A: "Auditability and portability — one place to check every input, one cell to re-point the whole model to a new ticker, and the calc engine keeps working even if the data connection drops."

Practise free

The formula *thinking* transfers completely without a licence: - **Python `yfinance`** — `yf.Ticker("INFY.NS").financials` and `.history(period="5y")` reproduce BDH/CIQ pulls for Indian tickers (`.NS` = NSE). Build the exact `Data_Raw` layout in pandas and export to Excel. - **Screener.in export** — the "Export to Excel" button on any company gives you 10 years of financials in one sheet; treat it as your CapIQ dump and build the clean tab on top. - **`nsepy` / NSE bhavcopy** — free historical prices for the BDH drill. - **`=STOCKHISTORY()`** in Microsoft 365 Excel — a genuine free BDH analogue for prices: `=STOCKHISTORY("INFY","1/1/2021",,2,0,0,1)`. Drill: replicate the Infosys revenue/EBITDA table above from a Screener.in export, then rebuild it in pandas so you understand what the plugin does under the hood.

Building a Comparable Companies (Comps) Analysis

What you'll be able to do

Build a trading-comps table from scratch: choose a defensible peer set, pull the raw inputs (market cap, net debt → enterprise value, revenue, EBITDA, EBIT, net income, shares, EPS), compute the multiples (EV/EBITDA, EV/Sales, EV/EBIT, P/E, PEG), calendarise so mismatched fiscal years are comparable, strip one-offs so the numbers are clean, take the median, and apply it to your target to imply a value and a football-field range. Worked end-to-end on Indian IT services with Infosys as the target.

The drill — step by step

1. Pick the peer set — this is 50% of the work. Comparable means *comparable*: same industry, similar business model, similar size, geography and margin profile. Use `EQS` /BICS on Bloomberg or CapIQ Screening: BICS "IT Consulting & Services", country India, market cap band. For Infosys the honest peer set is TCS, Wipro, HCLTech, Tech Mahindra, LTIMindtree. You could add Accenture/Cognizant as global reference points, but flag them — different scale and currency. **Write down your inclusion rule**; you will be asked to defend every name.

2. Pull the raw inputs into a clean grid (from your `Data_Raw` tab, Chapter 2). Per company you need:

- Share price and diluted shares → **Equity value (market cap)**
- Total debt + preferred + minority interest − cash & equivalents → **Net debt**
- **Enterprise Value (EV) = Equity value + Net debt** (+ minorities + preferred)
- Revenue, EBITDA, EBIT, Net income, EPS — both **LTM** (last twelve months) and **forward** (NTM / FY+1 consensus, `BEST_` / `IQ_NTM`)

3. Compute the multiples. The iron rule: **numerator and denominator must belong to the same claimants.**

Multiple	Formula	Whose metric
EV/EBITDA	EV ÷ EBITDA	capital-structure-neutral, pre-D&A
EV/EBIT	EV ÷ EBIT	after D&A, still pre-financing
EV/Sales	EV ÷ Revenue	for low/negative-margin names
P/E	Price ÷ EPS (or Equity ÷ Net income)	equity holders only
PEG	P/E ÷ EPS growth %	P/E normalised for growth

EV pairs with pre-interest metrics (EBITDA, EBIT, Sales) because EV represents *all* capital providers. **P/E** pairs with net income/EPS because both are *after* interest — equity-only. Never put EV over net income, or price over EBITDA.

4. Calendarise. Infosys, TCS, Wipro run **March year-ends**; Accenture runs **August**, Cognizant **December**. Comparing raw FY figures mixes periods. Fix by converting everyone to LTM, or calendarising to a common calendar year: e.g. CY = (months of FYn in the calendar year × FYn) + (remaining months × FYn+1), weighted.

For a purely Indian peer set with aligned March ends this is mostly moot — which is itself a reason to keep the set domestic.

5. Adjust for one-offs (clean the metric). Strip non-recurring items so multiples reflect underlying earnings: remove litigation settlements, restructuring, one-time visa/impairment charges, gains on asset sales; normalise a tax rate if a period had a one-off tax credit. Document each adjustment in a footnote row. A peer that took a big restructuring charge will show a distorted EBIT — un-distort it or its multiple lies.

6. Summarise — median, not just mean. Compute **min, 25th percentile, median, mean, 75th, max** for each multiple down the peer column. **Lead with the median** — it's robust to one outlier; the mean gets dragged by a single 40x name.

7. Imply the target's value. Apply the peer median multiple to the target's own metric:

- Implied EV = peer median EV/EBITDA × target EBITDA
- Implied Equity = Implied EV − target net debt
- Implied price = Implied equity ÷ target diluted shares
- Cross-check with P/E: Implied price = peer median P/E × target EPS

8. Football field. Plot each methodology's implied per-share range (25th–75th percentile of each multiple) as a horizontal bar; the overlap is your defensible value range, next to the current price.

Worked example — Infosys target, illustrative LTM figures:

Company	EV (₹ cr)	EBITDA (₹ cr)	EV/EBITDA	Net income (₹ cr)	P/E
TCS	14,20,000	62,000	22.9x	46,000	30.9x
Wipro	2,60,000	22,000	11.8x	13,100	19.8x
HCLTech	3,90,000	34,000	11.5x	15,700	24.8x
Tech Mahindra	1,30,000	9,500	13.7x	4,300	30.2x
LTIMindtree	1,60,000	8,200	19.5x	4,600	34.8x
Median			**13.7x**		**30.2x**

Apply to Infosys (EBITDA ≈ ₹38,900 cr, net debt ≈ −₹30,000 cr net cash, ~415 cr shares, EPS ≈ ₹66): - Implied EV = 13.7 × 38,900 ≈ **₹5,32,930 cr** - Implied equity = 5,32,930 + 30,000 (net cash adds back) ≈ **₹5,62,930 cr** - Implied price = 5,62,930 cr ÷ 415 cr ≈ **₹1,357/sh** - P/E cross-check: 30.2 × 66 ≈ **₹1,993/sh** → the spread tells you which multiple the market is really paying and that Infosys screens richer on P/E than EV/EBITDA vs. this set.

The output

A one-page comps sheet: peer rows with EV build-up and each multiple; a stats block (min/25th/median/mean/75th/max) per multiple; a target-implied-value block showing EV → equity → per-share for EV/EBITDA and P/E; and a football-field chart with the implied ranges against the live price. Footnotes list peer-selection rationale and every one-off adjustment.

Checks & gotchas

- **EV must be consistent** — always subtract cash and add debt, minorities and preferred the same way for every peer, or the ranking is noise.
- **Net cash companies** (most Indian IT) have EV < market cap; adding back net cash *raises* implied equity. Get the sign right.
- **LTM vs. forward mismatch** — never compare one peer's forward multiple to another's trailing. Pick one basis for the whole table.
- **Diluted, not basic shares** — include options/RSUs; treasury-stock method.
- **Outliers** — a high-growth or newly-listed peer can carry a 40x multiple; that's why you lead with the median and eyeball the range.
- **Currency** — don't mix an Accenture USD EV/EBITDA into an INR table without noting the multiple is currency-neutral but the size context isn't.
- **Garbage peers = garbage output.** The most common mistake is a lazy peer set. Comps are only as good as comparability.

Interview drill

Q: "Why EV/EBITDA over P/E?" A: "EV/EBITDA is capital-structure-neutral and pre-D&A, so it compares operating value across companies with different debt loads and depreciation policies — cleaner for cross-company comparison and for buyers who'll refinance the target. P/E is distorted by leverage and tax. I'd use both and explain divergences."

Q: "Your target trades at a discount to the peer median EV/EBITDA — buy signal?" A: "Not automatically. A discount can be justified — slower growth, lower margins, weaker returns on capital, governance or client-concentration risk. I'd check whether the discount is explained by fundamentals before calling it cheap. Comps tell you *where* it trades, not *why*."

Q: "How do you handle mismatched fiscal year-ends in a comps set?" A: "Calendarise — convert everyone to LTM or to a common calendar year by weighting the overlapping fiscal periods — so I'm comparing the same window. For an all-Indian March-end set it's usually moot, which is one reason I prefer a domestic peer group."

Practise free

- **Screener.in / TIKR / stockanalysis.com** — pull EV, EBITDA, net income, P/E for TCS, Wipro, HCLTech, Tech Mahindra, LTIMindtree and Infosys and build the exact grid above by hand.
- **Tijori Finance** — free peer-comparison views that mirror an RV screen.
- **Python** — `yfinance` `.info` gives `enterpriseValue`, `enterpriseToEbitda`, `trailingPE`; loop over the peer tickers, `pd.DataFrame`, compute median, imply Infosys value. This is the single best rep: it forces you to build EV yourself and see where net cash flips the sign. Drill target: reproduce the median EV/EBITDA and the implied Infosys price, then write the two-line rationale for why Infosys deserves a premium or discount to it.

Building a Precedent Transactions Analysis

What you'll be able to do

Build a precedent-transactions (deal comps) valuation: source relevant M&A deals from CapIQ Transactions or Mergermarket, screen them to a defensible set, pull the transaction value and target financials at the time of each deal, compute transaction multiples (EV/EBITDA, EV/Sales, EV/EBIT), read the control premium implied, adjust for timing/synergies, and apply the median to your target to imply an acquisition value. You'll know exactly why precedent multiples sit *above* trading comps and when to trust them. Worked on an Indian IT-services target.

The drill — step by step

1. Understand what you're building. Trading comps (Chapter 3) value a company as an independent public entity. Precedent transactions value it as an *acquisition target* — what real buyers actually paid to gain control. Deal multiples embed a **control premium** (typically 20–40% over the undisturbed price) and often **synergies**, so they run higher. Use precedents when the question is "what would someone pay to buy this?"

2. Source the deals. In **CapIQ → Transactions** (or **Mergermarket**), screen the M&A database on:

- **Industry** of the target (BICS/GICS = IT Services)
- **Geography** (India targets; add global IT-services deals as reference, flagged)
- **Date** — last 3–5 years only; older deals reflect a different rate/valuation regime
- **Deal size** band comparable to your target
- **Stake** — control deals (>50%, ideally 100%); exclude minority stakes, they don't carry a control premium
- **Deal status** — completed (or announced), not rumoured/withdrawn
- **Consideration** — note cash vs. stock (stock deals can inflate headline value)

Bloomberg equivalent: `MA <GO>` (M&A / deal search) and `EVTS`. Each deal record gives announced date, target, acquirer, deal value, stake, and often the target's LTM financials at announcement.

3. Pull the inputs per deal. For each transaction you need the **Transaction Enterprise Value (TEV)** — offer equity value + target net debt assumed — and the **target's LTM financials at the announcement date** (revenue, EBITDA, EBIT). Critical: use the financials as they were *at the time of the deal*, not today's. CapIQ Transactions usually stores these; otherwise pull the target's LTM as of the announcement quarter.

4. Compute transaction multiples.

Multiple	Formula
EV/EBITDA	TEV ÷ target LTM EBITDA at announcement
EV/Sales	TEV ÷ target LTM revenue
EV/EBIT	TEV ÷ target LTM EBIT

Same claimant-consistency rule as trading comps — TEV over pre-interest metrics.

5. Read the control premium. Where the target was public, premium = (offer price per share ÷ undisturbed price ~1 day/1 month before announcement) − 1. Record it; the median premium is itself a data point ("deals in this space cleared at ~25–30% premia").

6. Adjust for timing and synergies. - **Timing / regime:** a 2021 deal struck at peak multiples over-states today's fair value in a higher-rate 2026 environment. Weight recent deals more; note the vintage. - **Synergies:** strategic buyers pay up for cost/revenue synergies embedded in the price. That's why a *strategic* precedent multiple can overshoot standalone value — flag strategic vs. financial-sponsor deals; sponsor deals are cleaner "no-synergy" reads. - **Deal structure:** all-stock deals at a frothy acquirer price inflate headline TEV; note consideration mix.

7. Summarise and apply. Take **min/25th/median/75th/max** of each transaction multiple; **lead with the median**. Apply to the target's *current* LTM metric:

- Implied TEV = median deal EV/EBITDA × target LTM EBITDA
- Implied equity = Implied TEV − target net debt (+ net cash)
- Implied price = Implied equity ÷ diluted shares

Worked example — a mid-cap Indian IT-services target, illustrative deals:

Announced	Target	Acquirer	TEV (₹ cr)	LTM EBITDA (₹ cr)	EV/EBITDA	Premium
2022	Mid-cap IT co A	Global SI	12,000	750	16.0x	32%
2023	Digital eng. co B	PE sponsor	6,500	480	13.5x	26%
2021	ER&D services C	Strategic	9,000	500	18.0x	40%
2024	BPM/IT co D	Global SI	7,200	540	13.3x	24%
Median					**14.75x**	**29%**

Apply the median 14.75x to a target with LTM EBITDA ≈ ₹900 cr and net cash ≈ ₹500 cr, ~50 cr diluted shares: - Implied TEV = 14.75 × 900 ≈ **₹13,275 cr** - Implied equity = 13,275 + 500 ≈ **₹13,775 cr** - Implied price = 13,775 cr ÷ 50 cr ≈ **₹276/sh**

Note this sits above where trading comps (say median 11–12x) would land — that gap is roughly the control premium plus synergy value.

The output

A one-page deal-comps sheet: one row per transaction (date, target, acquirer, stake, consideration, TEV, target LTM metric, each multiple, premium); a stats block leading with the median; a "vintage" column so recency is visible; and an implied-value block (TEV → equity → per-share) applied to your target's current metrics. Footnotes flag strategic-vs-sponsor and any all-stock deals. Usually shown as the *top* bar of the football field, above trading comps.

Checks & gotchas

- **Point-in-time financials.** Use the target's EBITDA *as of the deal announcement*, not its latest — the multiple is a ratio of the price paid to what the buyer bought then. Mixing today's EBITDA with an old price is the classic error.
- **Stale deals.** Pre-2021 multiples reflect near-zero rates; don't average a 2019 deal against 2025 without weighting for regime.

- **Strategic vs. financial buyer.** Strategic multiples carry synergies; sponsor deals don't. Blending them without a flag over-states standalone value.
- **Minority vs. control.** A 15% stake purchase has no control premium — exclude it or you'll understate.
- **Consideration mix.** All-stock deals can inflate TEV if the acquirer's stock was rich at signing.
- **Thin data.** M&A in a niche can give you only 3–4 usable deals — say so; a precedent set of two is a range, not a valuation.
- **Deal value ≠ equity value.** TEV includes assumed net debt; don't quote it as the equity cheque.

Interview drill

Q: "Why do precedent transactions usually give a higher value than trading comps?" A: "Because deal prices include a control premium — buyers pay above the undisturbed market price to acquire control — and often synergies a strategic buyer expects to realise. Trading comps value the company as-is in the public market; precedents value it as an acquisition. The gap is roughly premium plus synergy."

Q: "Which financials do you use for the target in a precedent multiple — current or at the time of the deal?" A: "At the time of the deal. The multiple is price-paid over what the buyer was buying, so I use the target's LTM figures as of the announcement date. Using today's financials against an old price gives a meaningless ratio."

Q: "When are precedent transactions less reliable than comps?" A: "When deals are stale — struck in a very different rate or valuation regime — when the sample is tiny, or when the set mixes strategic and sponsor buyers with very different synergy assumptions. Deal data is also lumpy and lags; trading comps update daily. I'd caveat the range accordingly."

Practise free

No paid deal database? Reconstruct precedents from public sources: - **BSE/NSE announcements & SEBI SAST filings** — open-offer documents disclose offer price, stake and undisturbed price → compute the control premium directly. - **Company press releases / investor decks** — acquirers publish deal value and target revenue/EBITDA; capture TEV and the metric. - **VCCEdge / Tracxn (free tiers), news archives (Mint, ET, Business Standard, Reuters)** — India M&A deal terms. - **Mergermarket / CapIQ Transactions** if you have campus/library access. Drill: pick five real Indian IT/BPO acquisitions from the last four years, dig the TEV and target LTM EBITDA out of the press releases, build the exact table above, take the median, and imply a value for a listed mid-cap target — then compare it to where trading comps put the same company and explain the gap.

Building a Full DCF from Terminal Data

What you'll be able to do

Pull a company's historicals off a terminal (Bloomberg/CapIQ or a free proxy), forecast unlevered free cash flow (FCFF) for five years, build a WACC bottom-up using CAPM, layer in a terminal value two ways (Gordon growth and exit multiple), discount everything, walk enterprise value down to a per-share equity value, and pressure-test it with a 2-way sensitivity grid and a football-field chart. You will carry one real-ish company — call it **Bharat Consumer Ltd (BCL)** — all the way to a rupee-precise fair value, and you'll know how to spot when your own model is quietly cheating.

The drill — step by step

Step 1 — Pull the raw financials. On Bloomberg, `BCL IN Equity <GO>`, then `FA <GO>` (Financial Analysis). Tab to Income Statement and Cash Flow, set periodicity Annual, and export to Excel with the red `Export` button (or `=BDH("BCL IN Equity","SALES_REV_TURN","2021","2025")` in the Excel add-in). On CapIQ it's the "Financials" template. Free proxy: Screener.in export, or Tijori/annual report PDFs. Grab five years so you have growth rates and margins to anchor forecasts.

BCL FY25 actuals (₹ cr): Revenue 12,000, EBITDA 2,160 (18.0%), D&A 360, EBIT 1,800, tax rate 25.17% (India new regime + surcharge/cess), Capex 420, and change in working capital that consumed ₹150 cr as sales grew.

Step 2 — Forecast the revenue driver. Don't forecast 15 lines; forecast the driver and let margins do the rest. BCL grew revenue ~11% historically; I taper it: 11% → 10% → 9% → 8% → 7%. Hold EBITDA margin flat at 18% (state this assumption explicitly — margin expansion is where models lie). D&A ≈ 3% of sales, capex ≈ 3.5% of sales, working capital drag ≈ 1.25% of incremental sales.

Step 3 — Build FCFF, year by year. The formula, memorise it:

```
FCFF = EBIT × (1 − tax) + D&A − Capex − ΔNWC
```

Year 1 (FY26): Revenue = 12,000 × 1.11 = 13,320. EBITDA = 2,397.6. D&A = 399.6. EBIT = 1,998. NOPAT = 1,998 × (1−0.2517) = 1,495.1. + D&A 399.6 − Capex (3.5%×13,320 = 466.2) − ΔNWC (1.25% × 1,320 = 16.5) = **FCFF ₹1,412.0 cr**.

Roll the same engine forward:

₹ cr	FY26	FY27	FY28	FY29	FY30
Revenue	13,320	14,652	15,971	17,249	18,456
EBIT	1,998.0	2,197.8	2,395.7	2,587.3	2,768.4
NOPAT	1,495.1	1,644.6	1,792.7	1,936.1	2,071.6
+ D&A	399.6	439.6	479.1	517.5	553.7
− Capex	466.2	512.8	559.0	603.7	646.0
− ΔNWC	16.5	16.7	16.5	16.0	15.1
FCFF	**1,412.0**	**1,554.7**	**1,696.3**	**1,833.9**	**1,964.2**

Step 4 — Build WACC from CAPM. Risk-free: `GIND10YR Index <GO>` → 10-yr G-sec ≈ **6.9%**. Beta: on Bloomberg `BCL IN Equity BETA <GO>` gives raw and adjusted beta; use adjusted (Blume) ≈ **0.85** (or lever an industry beta). Equity risk premium for India: Damodaran's mature-market + country risk ≈ **7.5–8.0%**; use 7.5%.

```
Cost of equity = Rf + β × ERP = 6.9% + 0.85 × 7.5% = 13.28%
```

Cost of debt: BCL's coupon / rating spread ≈ 8.5% pre-tax, after-tax = 8.5% × (1−0.2517) = 6.36%. Capital weights at market: equity ₹28,000 cr, debt ₹4,000 cr → E/V = 87.5%, D/V = 12.5%.

```
WACC = 0.875 × 13.28% + 0.125 × 6.36% = 11.62% + 0.795% = 12.42%
```

Round to **12.4%**.

Step 5 — Terminal value, two ways. Gordon: pick g = 5.0% (below nominal GDP ~10.5%, sanity intact).

```
TV = FCFF_FY30 × (1+g) / (WACC − g) = 1,964.2 × 1.05 / (0.124 − 0.05)
   = 2,062.4 / 0.074 = ₹27,870 cr
```

Exit multiple cross-check: apply 12× EV/EBITDA to FY30 EBITDA (18% × 18,456 = 3,322): TV = ₹39,864 cr. The two disagree — that's the point; the Gordon TV implies a cheaper business. I'll run the DCF on Gordon and show exit as a bound.

Step 6 — Discount. Discount factor = 1/(1.124)^t (mid-year convention optional; here year-end).

	FY26	FY27	FY28	FY29	FY30	TV
CF	1,412.0	1,554.7	1,696.3	1,833.9	1,964.2	27,870
DF @12.4%	0.8897	0.7916	0.7043	0.6266	0.5575	0.5575
PV	1,256.3	1,230.7	1,194.7	1,149.1	1,095.1	15,538.5

Sum of PV(explicit FCFF) = ₹5,925.9 cr. PV(TV) = ₹15,538.5 cr. **Enterprise Value = ₹21,464 cr.**

Step 7 — EV-to-equity bridge.

```
Equity value = EV − Net debt − Minority interest − Preferred + Investments/associates
```

BCL: EV 21,464 − Net debt (debt 4,000 − cash 900 = 3,100) − minority 200 + associate investments 300 = **₹18,464 cr**. Shares outstanding = 100 cr. **Fair value = ₹184.6 / share.** Current price ₹280 → the DCF says overvalued ~34%; you'd flag that, not force the assumptions to fit.

The output

A one-screen model block: the FCFF build (table above), a WACC box (Rf 6.9%, β 0.85, ERP 7.5%, Ke 13.28%, Kd(at) 6.36%, weights 87.5/12.5, WACC 12.4%), the discounting strip, and the bridge:

```
EV                      21,464
 − Net debt             (3,100)
 − Minority               (200)
 + Associates              300
= Equity value          18,464
÷ Shares (cr)              100
= Value / share       ₹ 184.6
```

TV as % of EV = 15,538.5 / 21,464 = **72.4%**. Implied exit EV/EBITDA on the Gordon TV = 27,870 / 3,322 = **8.4×** — internally consistent and *below* the 12× comps multiple, so the DCF is conservative.

Checks & gotchas

- **TV should be 60–80% of EV.** 72% is normal. Above ~85% means your explicit window is too short or g too high.
- **Implied terminal growth vs implied exit multiple** must both be sane. Back out each from the other and eyeball them.
- **g < WACC always**, and g ≤ long-run nominal GDP. g of 8% in a 12.4% WACC world quietly doubles your TV.
- **Consistency of nominal terms**: nominal cash flows → nominal WACC (India, don't mix a real Rf with nominal growth).
- **Tax rate**: use the marginal/statutory rate for NOPAT, not the accounting effective rate distorted by one-offs.
- **Don't double-count**: FCFF is pre-financing, so discount at WACC and *never* subtract interest in the cash flow. Subtract debt only in the bridge.
- **Mid-year convention** lifts value ~6% (half a period less discounting); pick one and disclose it.
- **Circularity**: market-value weights depend on the equity value you're solving for. Iterate once or fix weights at target capital structure.

Interview drill

Q: Why FCFF discounted at WACC and not FCFE at cost of equity? FCFF is capital-structure-neutral — it's the cash to all providers, so you discount at the blended cost of all capital (WACC) and get enterprise value.

FCFE is post-debt cash to equity, discounted at Ke, giving equity value directly. FCFF is preferred when leverage is changing because WACC is more stable than a moving Ke; you avoid re-forecasting the debt schedule inside the cash flow.

Q: Your terminal value is 72% of EV — is the model useless? No, that's structurally normal because a going concern's value is dominated by cash flows beyond any five-year window. What matters is that the terminal assumptions are defensible: g of 5% is below nominal GDP, and the implied exit multiple of 8.4× is below trading comps of 12×, so I'm not smuggling in optimism through the terminal — if anything I'm conservative.

Q: Move ERP from 7.5% to 8.5% — direction and rough magnitude? Ke rises ~0.85%, WACC rises ~0.74%, so the denominator (WACC − g) widens and value falls sharply because TV is convex in the spread. A move from 12.4% to 13.1% WACC roughly cuts fair value from ₹185 toward ₹160 — a ~13% hit from a 1-point ERP change, which is exactly why the sensitivity table matters.

Practise free

You don't need a live terminal. Pull five years of financials from **Screener.in** (free export to Excel) or the company's annual report; use **GIND10YR** substitute from RBI/worldgovernmentbonds.com for the risk-free rate; get **beta** by regressing 2 years of weekly stock returns against Nifty in Excel (`=SLOPE(stock_returns, index_returns)`) — that literally *is* what the terminal's BETA function does. Take **ERP** from Damodaran's free country-risk-premium spreadsheet (updated each January). Build the whole FCFF → WACC → TV → bridge in one Excel tab, then add a two-input `Data Table` (Data ▸ What-If ▸ Data Table) with WACC across the top and g down the side to reproduce the sensitivity grid and football field for free.

Building an LBO Model (Sources/Uses, Debt Schedule, Returns)

What you'll be able to do

Take a target, set an entry enterprise value off an EBITDA multiple, build a Sources & Uses table that funds the deal, layer a realistic debt stack (Term Loan A/B + a revolver + optionally mezzanine), forecast EBITDA and free cash flow, run a cash sweep that pays debt down year by year, exit at a multiple, and compute the two numbers a PE interviewer actually cares about — **IRR** and **MOIC** — then stress them on leverage and exit. You'll carry one target, **Deccan Components Ltd (DCL)**, from a ₹1,000 cr entry to a five-year return, to the rupee.

The drill — step by step

Step 1 — Set the entry. DCL LTM EBITDA = ₹150 cr. Entry multiple = 8.0× → **Entry EV = ₹1,200 cr**. Assume no existing debt acquired (cash-free, debt-free deal), so equity purchase price = EV = 1,200. This is the "purchase enterprise value" line.

Step 2 — Decide leverage. Sponsors lever to what cash flow can service. Total debt = 4.5× EBITDA = ₹675 cr, split:

Tranche	× EBITDA	₹ cr	Rate	Amort
Term Loan A	2.0×	300	9.5%	10%/yr mandatory
Term Loan B	2.0×	300	10.5%	1%/yr, bullet
Revolver	—	0 drawn	9.0%	as needed
Total debt	**4.5×**	**675**		

Step 3 — Sources & Uses. Uses = what you must pay for. Sources = how you fund it. They must balance; the sponsor equity is the plug.

```
USES                          SOURCES
Purchase EV         1,200     Term Loan A          300
Transaction fees       30     Term Loan B          300
Financing fees         15     Revolver               0
                             Sponsor equity (plug) 645
TOTAL USES          1,245     TOTAL SOURCES       1,245
```

Sponsor equity = 1,245 − 675 = **₹570… wait, plug to balance: 1,245 − 600 (TLA+TLB) = ₹645 cr.** (Note the ₹45 cr of fees are funded by equity — that's why equity is 645, not 570.) Equity as % of total capitalisation = 645/1,245 = 51.8%.

Step 4 — Project EBITDA and FCF. DCL grows revenue 8% with 25% EBITDA margin holding. Build the cash available for debt paydown:

```
Cash for sweep = EBITDA − cash interest − cash taxes − capex − ΔNWC − mandatory amort
```

Year 1: EBITDA 150 × 1.08 = 162. Interest: TLA 9.5%×300 = 28.5, TLB 10.5%×300 = 31.5 → 60.0. D&A 30, EBIT 132, less interest 60 = PBT 72, tax @25% = 18, so cash taxes 18. Capex = 6% of sales (sales ≈ 648) = 38.9. ΔNWC = 8. Mandatory amort TLA 10%×300 = 30, TLB 1%×300 = 3.

FCF before sweep = 162 − 60 − 18 − 38.9 − 8 = 37.1. Less mandatory amort 33 = **₹4.1 cr free for the cash sweep** in Year 1 (early years are tight — that's realistic).

Step 5 — Debt schedule with cash sweep. Each year: opening balance − mandatory amort − optional sweep = closing balance. Sweep hits the most expensive prepayable tranche first (usually TLB after TLA mandatory). Interest is computed on the average or opening balance (be consistent). Roll five years:

₹ cr	Y1	Y2	Y3	Y4	Y5
EBITDA	162.0	175.0	189.0	204.1	220.5
Cash interest	60.0	56.5	52.2	47.0	40.8
Cash taxes	18.0	21.4	25.1	29.2	33.7
Capex	38.9	42.0	45.4	49.0	52.9
ΔNWC	8.0	8.6	9.3	10.0	10.8
FCF pre-amort	37.1	46.5	57.0	68.9	82.3
Total debt paid	37.1	46.5	57.0	68.9	82.3
Debt closing	**637.9**	**591.4**	**534.4**	**465.5**	**383.2**

By Year 5, ₹675 cr of debt is down to **₹383.2 cr** — the deleveraging engine that drives PE returns.

Step 6 — Exit and returns. Exit at Year 5, EBITDA = ₹220.5 cr. Exit multiple — conservatively assume *multiple contraction* to 7.5× (you never underwrite to expansion):

```
Exit EV = 7.5 × 220.5 = 1,653.8
− Net debt at exit      (383.2)  [assume minimal cash]
= Exit equity value    1,270.6
```

MOIC = exit equity / entry equity = 1,270.6 / 645 = **1.97×**. IRR = MOIC^(1/5) − 1 = 1.97^0.2 − 1 = **14.5%**. Below the ~20% PE hurdle — so at 8.0× entry / 4.5× leverage / 7.5× exit, this deal is marginal. That's a *finding*, and knowing it beats a model that always says "great deal."

The output

The finished LBO is four linked blocks on one screen: Sources & Uses (balances at 1,245), the debt schedule (above), the FCF build, and a returns box:

```
Entry equity             645.0
Exit EV (7.5×)         1,653.8
Exit net debt            (383.2)
Exit equity           1,270.6
MOIC                     1.97×
IRR (5-yr)               14.5%
```

Return attribution (the "bridge" PE loves): of the ₹625.6 cr equity gain, **deleveraging** contributed 675 − 383 = ₹291.8 cr, **EBITDA growth** at constant multiple contributed (220.5−150)×7.5 = ₹528.8 cr, and **multiple change** contributed (7.5−8.0)×150 = −₹75 cr. That decomposition is the answer to "where do your returns come from?"

Checks & gotchas

- **Sources must equal Uses** to the rupee, every time, including fees. Fees are a Use funded by equity — forgetting them overstates returns.
- **The revolver is the backstop**: if FCF goes negative, you *draw* the revolver, you don't create negative debt paydown. Model a minimum cash balance.
- **Interest is circular** (interest depends on debt, debt depends on sweep, sweep depends on interest). Enable iterative calc or use a beginning-balance interest convention to break it cleanly.
- **Never underwrite multiple expansion.** Assume flat or contracting exit; if the deal only works on expansion, it doesn't work.
- **Cash taxes ≠ book taxes** once you have amortising fees and interest shields — keep a separate tax build.
- **MOIC and IRR must be consistent**: IRR ≈ MOIC^(1/years) − 1 only with a single entry and single exit and no interim dividends; a dividend recap changes both.
- **Sanity**: entry leverage 4.5× with EBITDA margin 25% and rising rates — check the interest coverage (EBITDA/interest) stays above ~2.5× or the deal is un-financeable.

Interview drill

Q: What are the three drivers of LBO returns? Deleveraging (FCF pays down debt, so equity grows even at a flat multiple), EBITDA growth (revenue growth plus margin expansion), and multiple arbitrage (buy low, sell high — but you never *underwrite* to this; you assume flat or lower exit). In my DCL model, deleveraging added ₹292 cr and EBITDA growth ₹529 cr, while the multiple contraction *cost* ₹75 cr — showing returns can survive a worse exit multiple if the operating story delivers.

Q: Why does more leverage boost IRR but raise risk? Higher debt shrinks the equity cheque, so the same absolute equity gain is a larger percentage return — that's the IRR lift. But it also raises fixed cash interest, thinning the FCF cushion; a small EBITDA miss can breach covenants or force a revolver draw, and equity is first-loss. Leverage amplifies both the upside and the probability of a zero.

Q: Walk me from entry EV to sponsor equity. Entry EV = entry multiple × LTM EBITDA. Add transaction and financing fees to get total uses. Fund it with the debt tranches you can raise given leverage capacity; sponsor equity is the plug that makes sources equal uses. So equity = total uses − total debt drawn, and it explicitly absorbs the fees.

Practise free

No LBO software needed — it's all Excel. Pull a mid-cap's EBITDA from **Screener.in**, pick an entry multiple from **comparable transactions** (news of recent PE deals, or trading comps as a floor). Build Sources & Uses, then a five-row debt schedule with a `=MIN(cash_available, debt_outstanding)` sweep and iterative calculation turned on (File ▸ Options ▸ Formulas ▸ Enable iterative). Compute IRR with `=RATE` or `=(exit/entry)^(1/n)-1`, and MOIC as a simple ratio. Add a `Data Table` sensitising IRR against entry leverage (rows) and exit multiple (columns) to reproduce the classic PE sensitivity grid. Rehearse the return-attribution bridge by hand — that verbal decomposition is what actually gets tested in PE interviews.

Writing the Equity Research Note / Initiation

What you'll be able to do

Turn a finished model into a sell-side **initiation report** that a desk would actually publish: a crisp one-line thesis, a rating and price target that *fall out of the model* (not the other way round), a business and industry section grounded in channel checks, an explicit forecast block, a blended DCF-plus-comps valuation, a risk register, catalysts, and a one-page summary — written in SEBI-compliant, non-promotional prose. You'll produce a short worked excerpt for **Bharat Consumer Ltd (BCL)** and know exactly what an Equirus/Nuvama/street-side analyst JD is asking for.

The drill — step by step

Step 1 — Write the thesis in one sentence, first. If you can't state why the stock is mispriced in a line, you don't have a note. BCL example:

> *"We initiate on BCL with a REDUCE and a ₹185 target (34% downside): the street is extrapolating 15% volume growth into a category that is structurally slowing to high-single-digits, and current multiples price in margin expansion the company has repeatedly guided against."*

Note the structure: rating, target, downside %, and the *variant perception* — what you see that the consensus doesn't.

Step 2 — Set rating and target from the model, transparently. The target must be reproducible. Blend your DCF (from chapter 5, ₹185) with a comps cross-check:

Method	Value/share	Weight
DCF (WACC 12.4%, g 5%)	₹185	60%
EV/EBITDA comps (11× FY27E)	₹196	25%
P/E comps (28× FY27E EPS)	₹178	15%
Blended target	**₹187**	100%

Round to **₹185**. Map to a rating band the desk uses (e.g. BUY >15% upside, ADD 5–15%, REDUCE −5 to −15%, SELL <−15%). At ₹280 vs ₹185, that's −34% → **REDUCE** (or SELL, per house bands).

Step 3 — Business & industry, grounded in channel checks. Don't rewrite the annual report. Explain the *unit economics* and the *industry structure*, then evidence it. Channel checks = the primary work: call 8–10 distributors, visit modern-trade shelves, check e-commerce pricing, talk to two ex-employees, read the last four earnings-call transcripts for a change in tone. Write it as evidence: "*Our checks across 12 distributors in 3 states indicate secondary sales decelerating to ~7% YoY, versus management's 12–14% primary-sales run-rate — implying channel inventory build.*" That specificity is the moat of a good note.

Step 4 — Forecasts, shown explicitly. Analysts publish their numbers so clients can disagree with a specific line. Present three years:

₹ cr	FY25A	FY26E	FY27E	FY28E
Revenue	12,000	13,320	14,652	15,971
growth %	—	11.0	10.0	9.0
EBITDA	2,160	2,398	2,637	2,875
margin %	18.0	18.0	18.0	18.0
PAT	1,290	1,433	1,578	1,720
EPS (₹)	12.9	14.3	15.8	17.2

State the two or three assumptions that matter (flat margin, tapering volume) and where you differ from consensus ("*our FY27 EPS is 6% below Bloomberg consensus of ₹16.8*").

Step 5 — Valuation section. Show the DCF assumptions box, the sensitivity grid (WACC × g), and the comps table so the target is auditable. Include a **football field** — DCF range, EV/EBITDA range, P/E range, 52-week range — with the current price line, so the reader sees where ₹280 sits versus every method.

Step 6 — Risks and catalysts. Risks are two-sided and specific: *upside risk* — faster rural recovery lifts volumes to 12%, taking the DCF to ₹215; *downside* — raw-material inflation compresses margin 150bps to ₹160. Catalysts with dates: Q1 results (Jul), monsoon data (Aug), a rumoured price hike. A note without dated catalysts is an opinion, not a trade.

Step 7 — One-pager and compliance. The front page is the whole note in one screen: rating, target, price, up/downside, market cap, the thesis in three bullets, a mini financial summary table, and the football field. Then the **SEBI RA disclosures**: registration number, analyst certification ("*views accurately reflect my personal views; no part of compensation was tied to the recommendation*"), holdings disclosure, and no forward-looking guarantee language. Prose must be **non-promotional**: no "must buy", no "guaranteed", no target framed as certainty — SEBI (Research Analysts) Regulations require balanced, evidence-based language.

The output

A worked excerpt of the initiation:

BCL IN | REDUCE | TP ₹185 (−34%) | CMP ₹280 | Mkt cap ₹28,000 cr

Thesis. We initiate coverage of Bharat Consumer with a REDUCE rating and a ₹185 12-month target. The market is pricing sustained mid-teens volume growth into a category our channel work shows decelerating to ~7%. At 28× FY27E EPS, the stock discounts margin expansion management has explicitly guided against.

Why now. (1) Secondary-sales checks across 12 distributors point to channel inventory build; (2) consensus FY27 EPS of ₹16.8 sits 6% above our ₹15.8; (3) the risk-reward is skewed — our bull case is ₹215 (+8% from a rural recovery) against a bear case of ₹160 (−15% on margin compression).

Valuation. DCF (WACC 12.4%, terminal g 5.0%) yields ₹185; an 11× FY27E EV/EBITDA cross-check gives ₹196. We weight the DCF 60% given the long-duration cash-flow profile. Football field below shows CMP above every method's central estimate.

Key risks to our call. Faster-than-expected rural volume recovery; benign input costs; a value-accretive acquisition.

RA Reg. No. INHxxxxxxxxx. Analyst certification and disclosures on p.12.

Checks & gotchas

- **The target must tie to the model.** If a reader rebuilds your DCF and can't get near ₹185, the note has no credibility. Keep the assumptions box in the report.
- **Rating consistency**: a REDUCE with a target 34% below price is right; a BUY with a target *below* the current price is an instant credibility kill — check the arithmetic.
- **Two-sided risks.** A note that only lists downside for a SELL (or only upside for a BUY) reads as a pitch, not research. SEBI-compliant means balanced.
- **No promissory language.** "Will", "guaranteed", "multibagger" are red flags for both compliance and quality.
- **Consensus context**: always state where you sit vs consensus — that's your variant perception, the reason the note exists.
- **Dated catalysts.** "Catalyst: better performance" is empty. Name the event and the month.
- **Disclosures are mandatory**, not optional garnish — RA registration, certification, holdings, and conflicts.

Interview drill

Q: What makes a research note different from a valuation model? The model produces a number; the note produces a *decision with a variant perception*. Its job is to identify where the market is wrong and why, evidence it with primary work like channel checks, and frame a risk-reward with catalysts — the target price is an output of that argument, not the argument itself. A great note could be right on the call even if the exact target is off by 5%.

Q: How do you blend DCF and comps into one target? They answer different questions — DCF is intrinsic and long-duration-sensitive, comps are relative and reflect current market sentiment. I weight toward DCF for stable cash-generative businesses and toward comps for cyclical or sentiment-driven names, then present both plus a football field so the client sees the range, not a false-precision single number. For BCL I used 60% DCF given its long, predictable cash flows.

Q: How do SEBI RA norms shape how you write? They require the recommendation to reflect genuine analysis, compensation not to be tied to the specific call, and full disclosure of holdings and conflicts, plus registration and analyst certification. Practically it forces balanced, evidence-based, non-promotional prose — two-sided risks, no guarantees — which, conveniently, is also what makes research credible to a professional reader.

Practise free

You don't need a Bloomberg terminal or a research license. Pick a listed company, build the model from **Screener.in** data (chapter 5's method), and do real **channel checks** for free — visit shops, check Amazon/Blinkit pricing, read the last four **earnings-call transcripts** (free on the company IR site or Screener's "Documents"), and note where management tone shifts. Read three or four **published initiations** (many are free on brokerage sites, Trendlyne, or research aggregators) purely to internalise the *structure and register* — thesis-first, evidence-dense, non-promotional. Then write a 2-page note with a one-pager and a football field (`Data Table` + a stacked bar in Excel). Study the SEBI (Research Analysts) Regulations disclosure list — it's public — and put a compliant disclosure block on your practice note so the habit is automatic.

The Modeling Test & Terminal Interview Drills + Cheat-Sheet

What you'll be able to do

Walk into a timed modelling test (the 60–90 minute build-a-3-statement-and-DCF-from-a-blank-sheet exercise) and know exactly what the reviewer is watching, how to structure for speed, which keyboard shortcuts save you ten minutes, and how to keep the model auditable. You'll also handle the rapid-fire terminal function quiz and the standing valuation Q&A, and you'll have a one-page cheat-sheet of every formula and function you need under time pressure.

The drill — step by step

Step 1 — Before you type, set the frame (2 min). Read the prompt, note the deliverable (usually: 3-statement + DCF, one or two scenarios, an answer to "is it cheap?"). Lay out tabs: `Assumptions`, `Model` (IS/BS/CF stacked), `DCF`, `Output`. Colour convention immediately — **blue = hardcoded input, black = formula, green = link to another sheet**. Reviewers grade this in the first 30 seconds.

Step 2 — Build the income statement top-down (10 min). One driver: revenue growth. Everything else as a % of revenue or a margin. Wire tax off PBT. Don't hardcode any calculated cell — if a number can be computed, compute it.

Step 3 — Balance sheet and the cash-flow bridge (20 min). Build working-capital via **days**: DSO, DIO, DPO drive receivables, inventory, payables. PP&E roll-forward: opening + capex − depreciation. Debt and equity from a simple schedule. Then the **cash flow statement links everything**: CFO from net income + D&A ± ΔNWC, CFI = −capex, CFF = debt/equity flows. Ending cash flows back to the balance sheet.

Step 4 — Make it balance (5 min). The whole test is really "does your balance sheet balance?" Build a check row: `Assets − Liabilities − Equity = 0` in every column, conditionally formatted red if non-zero. If it doesn't tie, 90% of the time it's the cash flow statement not fully feeding ending cash, or a working-capital sign error.

Step 5 — DCF on top (15 min). FCFF = EBIT×(1−t) + D&A − capex − ΔNWC (chapter 5). WACC box, terminal value (Gordon), discount, EV-to-equity bridge, per-share. Add a `Data Table` sensitivity (WACC × g).

Step 6 — Checks and a one-line answer (5 min). Turn on every check row. Then, in a cell or out loud, state the conclusion: "*Fair value ₹185 vs price ₹280 — overvalued, DCF-implied exit multiple 8.4× below comps 12×*." The reviewer wants a *view*, not just a number.

What they're actually watching

Dimension	Pass	Fail
Structure	Inputs separated, colour-coded, one driver	Hardcodes buried in formulas
Formulas	No plugs, everything links	Balancing figure typed in by hand
Checks	Balance check + cash tie, visible	No checks, silent errors
Speed	Shortcuts, no mouse for navigation	Formatting the sheet for 20 min
Judgement	States a view with a sanity check	Delivers a number with no context

Speed shortcuts (no-mouse modelling)

- `F2` edit cell, `F4` toggle absolute refs (`$A$1`), `Ctrl+;` today's date.
- `Alt+=` autosum. `Ctrl+Shift+arrow` select to edge. `Ctrl+D` / `Ctrl+R` fill down/right.
- `Alt,H,O,I` autofit column. `Alt,A,W,T` open Data Table. `Ctrl+[` trace precedent.
- `F9` recalc; `Ctrl+`` (grave) show all formulas for a fast audit.
- Name your WACC and tax cells so formulas read `=EBIT*(1-tax)` not `=D12*(1-$B$4)`.

Terminal / CapIQ function quiz (rapid fire)

Task	Bloomberg	CapIQ / free proxy
Price history to Excel	`=BDH( … )`	`=CIQ( … )` / Screener export
Current field (P/E, mkt cap)	`=BDP("TICK","PE_RATIO")`	`=CIQ("IQ_PE_EXCL")`
Company financials screen	FA <GO>	Financials template
Beta	BETA <GO>	`SLOPE()` regression (free)
Comparable companies	RV <GO>	Comps template
Relative valuation / football	EQRV <GO>	build manually
Debt / capital structure	DDIS <GO>	annual report
Estimates / consensus	EEO <GO> / EE <GO>	Trendlyne
Supply chain	SPLC <GO>	annual report
News run	CN <GO>	Google News

The output

A finished test artefact is: a colour-coded 3-statement model that balances in every column (visible check row of zeros), a DCF tab with a WACC box and a WACC×g sensitivity grid, and a one-line verdict. It looks disciplined — inputs in blue on their own tab, no orphaned hardcodes, checks green. And the one-pager below, which you can reconstruct from memory in the first two minutes of any test as your scaffold.

The one-page cheat-sheet

Free cash flow

```
FCFF = EBIT × (1 − t) + D&A − Capex − ΔNWC      → discount at WACC → EV
FCFE = FCFF − Interest×(1−t) + Net borrowing     → discount at Ke   → Equity
```

WACC & CAPM

```
Ke   = Rf + β × ERP
Kd   = pre-tax cost of debt × (1 − t)
WACC = (E/V)×Ke + (D/V)×Kd
```

Terminal value

```
Gordon:  TV = FCFF_n × (1+g) / (WACC − g)      [g < WACC, g ≤ nominal GDP]
Exit:    TV = EBITDA_n × exit multiple
Implied exit multiple = Gordon TV / EBITDA_n   (cross-check)
```

Bridge

```
Equity = EV − Net debt − Minority − Preferred + Associates
Per share = Equity / diluted shares
```

Multiples

```
EV/EBITDA, EV/EBIT, EV/Sales   (capital-structure neutral → use EV)
P/E, P/B, P/FCFE               (equity multiples → use price)
PEG = P/E ÷ growth
```

LBO

```
Sources = Uses (equity is the plug, fees are a Use)
MOIC = exit equity / entry equity
IRR  ≈ MOIC^(1/years) − 1      (single entry/exit, no interim CF)
Returns = deleveraging + EBITDA growth + multiple change
```

Best practice: blue input / black formula / green link · one driver, margins do the rest · no hardcoded plugs · balance-check row = 0 · TV = 60–80% of EV · nominal CF with nominal WACC · state a view.

Checks & gotchas

- **Balance-sheet imbalance** almost always = cash flow statement not fully wired or a working-capital sign flip. Fix the CF, don't plug the BS.

- **Circularity** (interest ↔ debt ↔ cash) — enable iterative calc *before* it errors, or use opening-balance interest.
- **Hardcoded plugs** are the single fastest way to fail; the reviewer traces one formula and finds it.
- **Formatting over substance** — don't spend 20 minutes on borders and a blank model. Function first, polish last.
- **No view** — a technically perfect model with no "cheap or dear?" conclusion still disappoints. Always land the plane.
- **Mixing real and nominal**, or effective vs marginal tax — silent value errors the reviewer will probe.

Interview drill

Q: Your model doesn't balance — walk me through debugging it. First, isolate the column where the check first goes non-zero — the error started that year. Then check the cash flow statement: does ending cash on the CF equal the cash line on the BS? Usually not, meaning a flow is missing or double-counted — most often a working-capital sign (an increase in receivables is a *use* of cash, negative in CFO) or capex not linked to both PP&E and CFI. I fix the source, never plug the balance sheet, because a plug hides the real error.

Q: You have 60 minutes for a 3-statement plus DCF — how do you spend it? Two minutes framing tabs and colour convention, ten on a driver-based income statement, twenty on the balance sheet and the cash-flow bridge, five making it balance with a visible check row, fifteen on the DCF and a sensitivity table, and the last five turning on checks and writing the one-line conclusion. I build ugly-but-linked first and only format if time remains — a balancing model with no borders beats a beautiful model that plugs.

Q: EV/EBITDA vs P/E — when each? EV multiples are capital-structure-neutral, so I use EV/EBITDA to compare companies with different leverage or to value the whole enterprise — it's above the interest line. P/E is an equity multiple, sensitive to leverage and tax, useful for comparing similar-capital-structure peers or when the market quotes the sector on earnings. For a leveraged or acquisitive name I lead with EV/EBITDA; for a stable financial or a consumer name the market prices on P/E, so I show both.

Practise free

Do the whole thing in plain Excel — no terminal required. Grab a company's five years from **Screener.in**, then rebuild the 3-statement model *from a blank sheet* against a timer (start at 90 minutes, work down to 60). Enforce the colour convention and a balance-check row every time. Replace terminal functions with their free equivalents: beta via `=SLOPE()` regression, risk-free from the RBI 10-year, ERP from Damodaran's free January dataset, comps built by hand from Screener. Rehearse the function quiz as flashcards (BDP/BDH, FA, RV, EQRV, BETA, EEO). Run a `Data Table` for the sensitivity grid. Time-box mercilessly and, at the buzzer, force yourself to write the one-line verdict — the muscle you're building is *finishing with a view under time pressure*, which is exactly what the real test grades.